Expt 4 Design of Low Pass Filter using Windowing Techniques

Exp 4.1: Design and Analysis of an FIR Low-Pass Filter Using the Rectangular Window Technique

Program:

```
clc;
clear;
close all;
%% Input Parameters
fp = 1000;      % Passband cutoff frequency (Hz)
fs = 4000;      % Sampling frequency (Hz)
N  = 50;        % Filter order
% Normalized cutoff frequency
wc = fp/(fs/2);
%% 1. FIR Low-Pass Filter using Rectangular Window
b_rect = fir1(N, wc, 'low', rectwin(N+1));
disp('Rectangular Window Coefficients:');
disp(b_rect');
%% 2. FIR Low-Pass Filter using Hamming Window
b_hamm = fir1(N, wc, 'low', hamming(N+1));
disp('Hamming Window Coefficients:');
disp(b_hamm');
%% 3. FIR Low-Pass Filter using Hann Window
b_hann = fir1(N, wc, 'low', hann(N+1));
disp('Hann Window Coefficients:');
disp(b_hann');
%% Frequency Response - Rectangular Window
figure(1);
freqz(b_rect,1,1024,fs);
title('FIR Low-Pass Filter using Rectangular Window');
```

```
%% Frequency Response - Hamming Window
figure(2);
freqz(b_hamm,1,1024,fs);
title('FIR Low-Pass Filter using Hamming Window');
%% Frequency Response - Hann Window
figure(3);
freqz(b_hann,1,1024,fs);
title('FIR Low-Pass Filter using Hann Window');
```

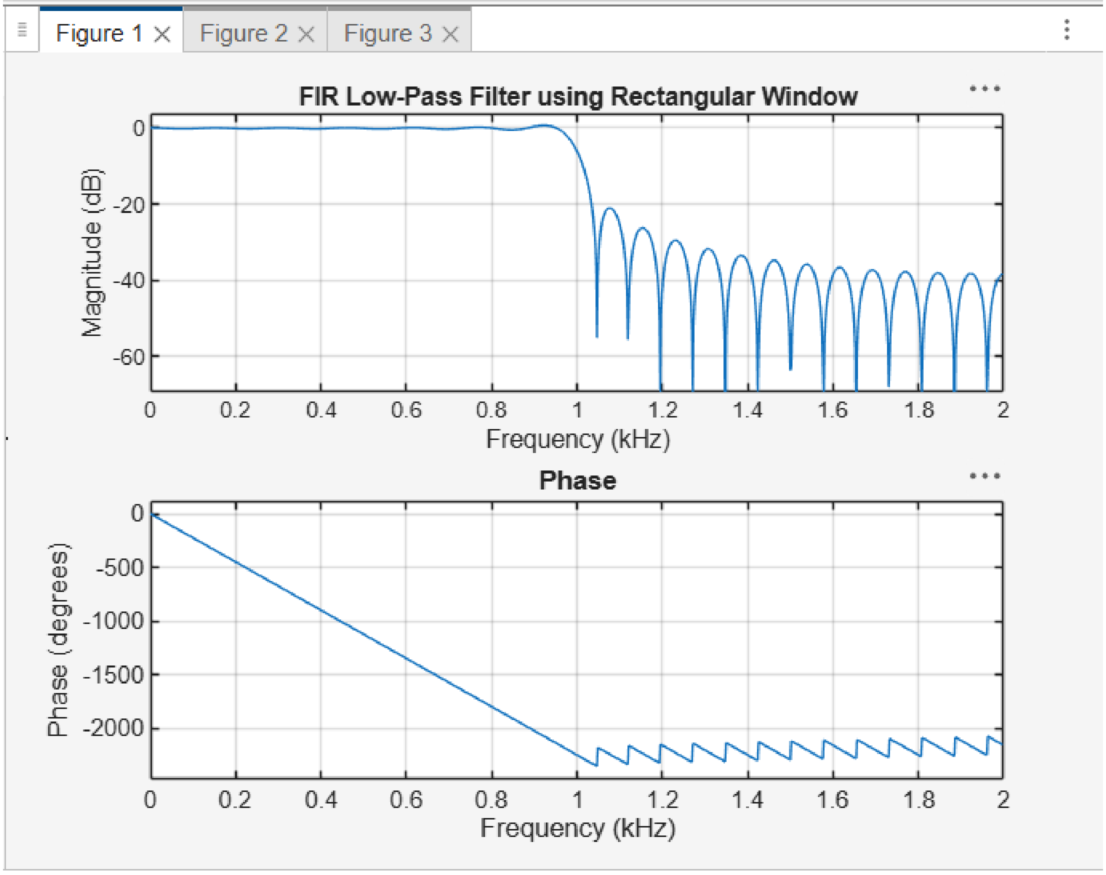


Figure 1
Figure 2
Figure 3
FIR Low-Pass Filter using Rectangular Window
Magnitude (dB)
0
-20
-40
-60
0 0.2 0.4 0.6 0.8 1 1.2 1.4 1.6 1.8 2
Frequency (kHz)
Phase
Phase (degrees)
0
-500
-1000
-1500
-2000
0 0.2 0.4 0.6 0.8 1 1.2 1.4 1.6 1.8 2
Frequency (kHz)

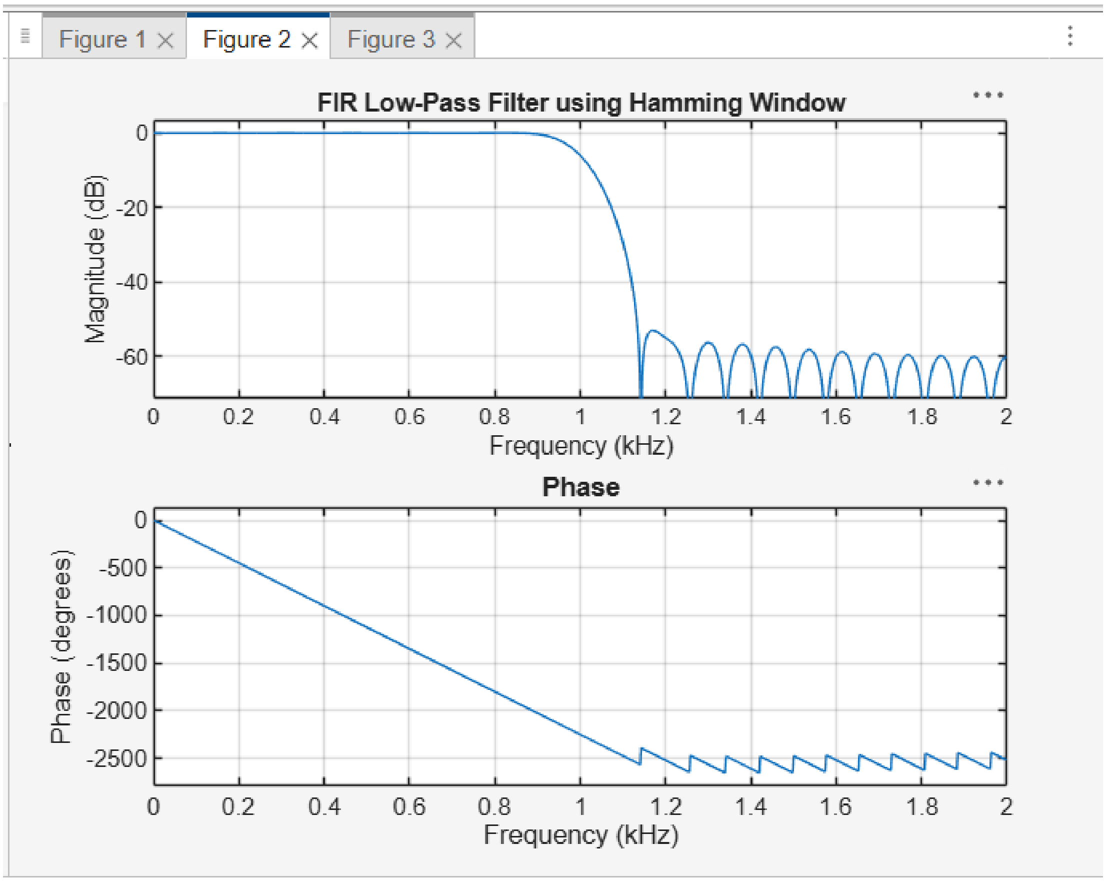
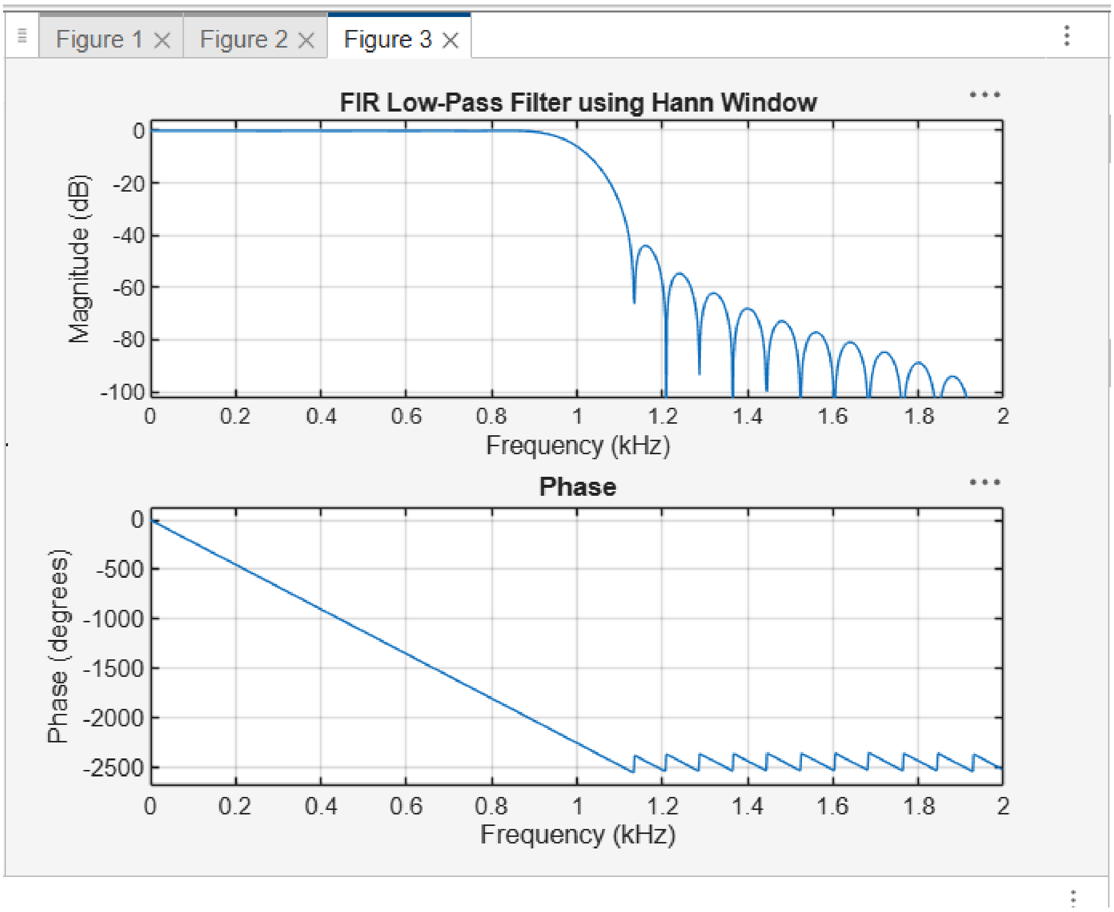


Figure 1
Figure 2
Figure 3
FIR Low-Pass Filter using Hamming Window
Magnitude (dB)
0
-20
-40
-60
0 0.2 0.4 0.6 0.8 1 1.2 1.4 1.6 1.8 2
Frequency (kHz)
Phase
Phase (degrees)
0
-500
-1000
-1500
-2000
-2500
0 0.2 0.4 0.6 0.8 1 1.2 1.4 1.6 1.8 2
Frequency (kHz)

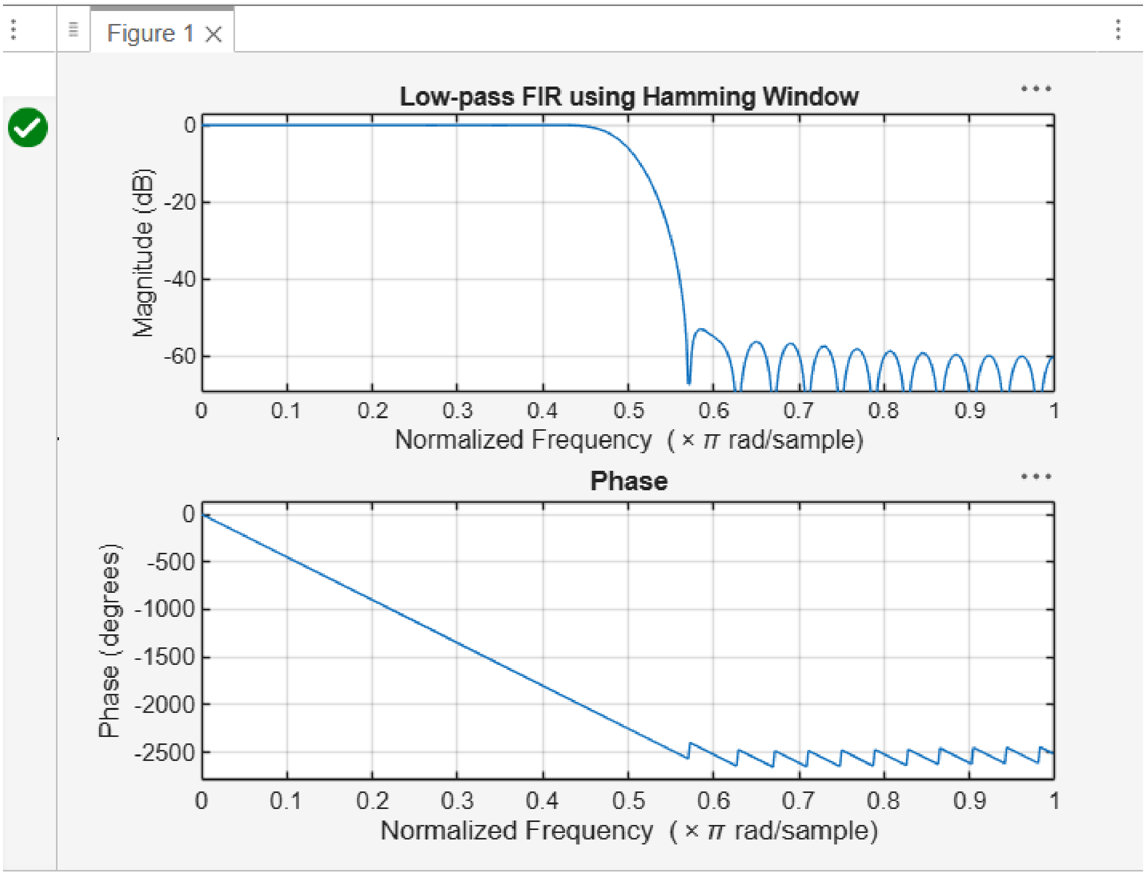


4.2. Design and Analysis of an FIR Low-Pass Filter Using the Hamming Window Technique

Program:

```
clc;
clear;
close all;
%% --- Input Parameters ---
fp = 1000; % Passband cutoff (Hz)
fs = 4000; % Sampling frequency (Hz)
N = 50; % Filter order (even number preferred)
wc = fp/(fs/2); % Normalized cutoff frequency (0 to 1)
%% --- FIR Low Pass using Hamming Window ---
b_hamm = fir1(N, wc, 'low', hamming(N+1));
disp('--- Hamming Window Coefficients ---');
```

```
disp(b_hamm');
%% --- Plot Frequency Responses ---
figure;
freqz(b_hamm,1);
title('Low-pass FIR using Hamming Window');
```



4.3. Design and Analysis of an FIR Low-Pass Filter Using the Hanning Window Technique

Program:

```
clc;
clear;
close all;
%% --- Input Parameters ---
fp = 1000; % Passband cutoff (Hz)
fs = 4000; % Sampling frequency (Hz)
N = 50; % Filter order (even number preferred)
```

```
wc = fp/(fs/2); % Normalized cutoff frequency (0 to 1)

%% --- FIR Low Pass using Hanning Window ---

b_hann = fir1(N, wc, 'low', hanning(N+1));

disp('--- Hanning Window Coefficients ---');

disp(b_hann');

%% --- Plot Frequency Responses ---

figure;

freqz(b_hann,1);

title('Low-pass FIR using Hanning Window');
```